

Quantum Mechanics

Lecture 1: Quantum Logic, Measurement, and Vector Spaces

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Chapter 1

Quantum Logic, Measurement, and Vector Spaces

Quantum mechanics is not classical mechanics with a little random noise added. Its unpredictability coexists with exact conservation and reversible evolution, yet the questions we may ask of a system cannot be separated from the physical means by which we ask them. The purpose of this first lecture is to make that difference visible before introducing the formal machinery. We shall move from the two-slit experiment and reversibility to Heisenberg's microscope, and only then ask what kind of mathematical object a quantum state must be.

1.1 The Contract of the Course

People who encounter these lectures on the Internet see the lecturer but not the audience, and the class can therefore look puzzling. It is neither a standard undergraduate course nor a graduate course. It belongs to Stanford's continuing-education program: people from the surrounding community, along with some people connected to the university, return to theoretical physics because they want the subject itself.

The age distribution makes the purpose unusually clear. Only a handful of the roughly one hundred people in the room are under forty; many are over seventy, a few over eighty, and earlier classes have even included people over ninety. The audience wants to reach the basic ideas quickly. That does not mean replacing physics by metaphors. We shall use equations, sometimes hard equations, but only as much machinery as is needed to get the ideas right. The aim is a theoretical minimum in the literal sense: efficient, straightforward, and exact enough to support the next step.

This lecture also belongs to a larger sequence of approximately six ten-lecture courses. Classical mechanics came first because it supplies the language of motion, energy, momentum, and time evolution. A course on quantum entanglement provided an earlier entrance to quantum ideas. The present course is nevertheless designed to be substantially self-contained. We begin with the deepest difference between the two subjects: classical and quantum physics do not organize possibilities according to the same logic.

1.2 Classical Randomness Is the Wrong Model

Quantum mechanics contains unpredictability, but of a very special kind. Einstein objected that the laws of nature should not amount to a throw of dice; Bohr's reply was that Einstein should not prescribe what nature may do. The familiar exchange can be misleading if it suggests that quantum theory simply inserts a random-number generator into Newton's laws. Let us construct that classical model first and see why it fails.

Imagine that the moon ordinarily obeys Newton's equation

$$\mathbf{F} = m\mathbf{a} \tag{1.1}$$

together with the law of gravity, but that every tenth of a second an independent throw of the dice gives it a small extra kick. One outcome pushes the moon slightly forward, another slightly backward, and most outcomes may do nothing. Its future orbit would become unpredictable even if its initial position and velocity were known exactly.

This randomness would leave a characteristic scar. Each kick would change the energy by a small positive or negative amount. The average change might vanish, but the accumulated energy would execute a random walk. Exact energy conservation would be lost.

Quantum evolution does not work that way. Prepare a closed system with a sharp energy and let it evolve under a time-independent Hamiltonian. A later energy measurement returns the same energy exactly, even though other measurement outcomes may be unpredictable.¹ Quantum uncertainty affects some questions while leaving the relevant conservation law exact. It cannot be represented by occasional violations of an otherwise classical equation.

1.3 Two Slits and the Failure of Classical Alternatives

The two-slit experiment gives the cleanest first view of the new logic. Photons, electrons, or neutrons may be used; even a bowling ball has a quantum wavelength in principle, but the corresponding effect is far too small to observe. Imagine a source so weak that particles leave one at a time, perhaps one every five minutes. Each particle passes a barrier and eventually produces one localized flash or one darkened spot on a screen.

Open only the upper slit. Repeated particles build a broad arrival profile, which we denote by $P_1(y)$. Close that slit and open only the lower one. The second experiment produces another broad profile $P_2(y)$, displaced on the screen. If the particles merely received independent classical kicks while passing the barrier, opening both slits would produce the pointwise sum

$$P_{\text{cl}}(y) = P_1(y) + P_2(y). \tag{1.2}$$

Every particle would use one route or the other, and two available routes could not make an already accessible point inaccessible.

The experiment gives an interference pattern instead. There are points y_0 for which

$$P_{12}(y_0) = 0, \quad P_1(y_0) > 0, \quad P_2(y_0) > 0. \tag{1.3}$$

¹Editorial clarification: The qualification about a closed system and a time-independent Hamiltonian states the standard condition implicit in the lecture's conservation argument.

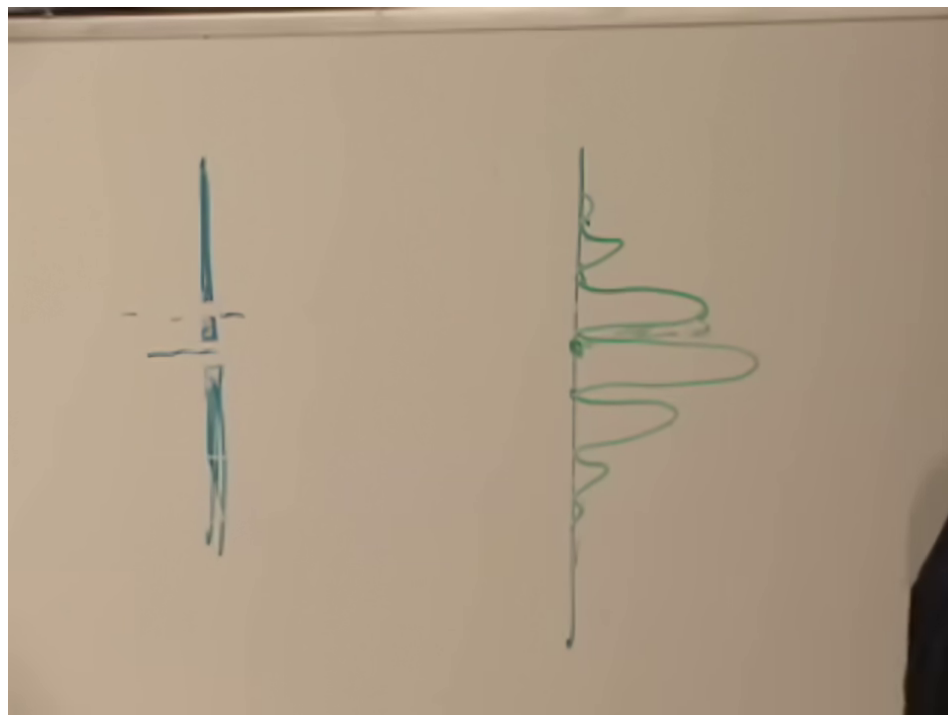


Figure 1.1: The completed two-slit interference profile on the board. The horizontal lobes represent arrival probability along the screen, including the nodes at which no particles arrive.

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Either slit by itself supplies arrivals at y_0 ; opening both removes them. The particles may be separated by minutes or years, so the pattern cannot be attributed to collisions among successive particles.

One can invent an elaborate memory hidden in the barrier or screen, but interference is far too general to be explained by a special mechanism built for this apparatus. The broad conclusion is unavoidable: the classical rule for mutually exclusive alternatives is not the fundamental rule. Later we shall express the replacement in terms of probability amplitudes, which may cancel before probabilities are formed. At this stage the observed nonadditivity in Eq. (1.3) is enough.

1.4 Reversibility and Conservation of Information

A second route to the same conclusion begins with deterministic motion in a finite state space. For a two-state coin, one reversible law may leave heads and tails unchanged,

$$H \mapsto H, \quad T \mapsto T, \quad (1.4)$$

while another may interchange them,

$$H \mapsto T, \quad T \mapsto H. \quad (1.5)$$

For a fanciful three-sided coin with heads, tails, and feet, consider

$$H \mapsto T, \quad T \mapsto F, \quad F \mapsto H. \quad (1.6)$$

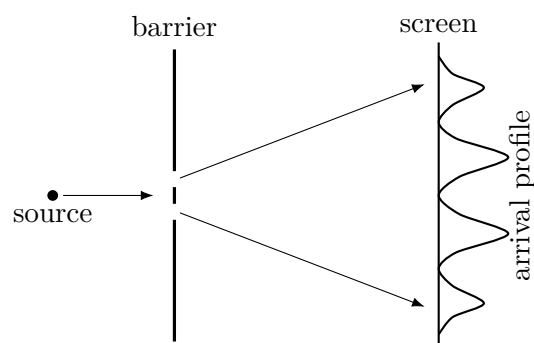


Figure 1.2: A clean reconstruction of the board geometry. The measured probability is built one detection at a time, yet it contains alternating maxima and nodes.

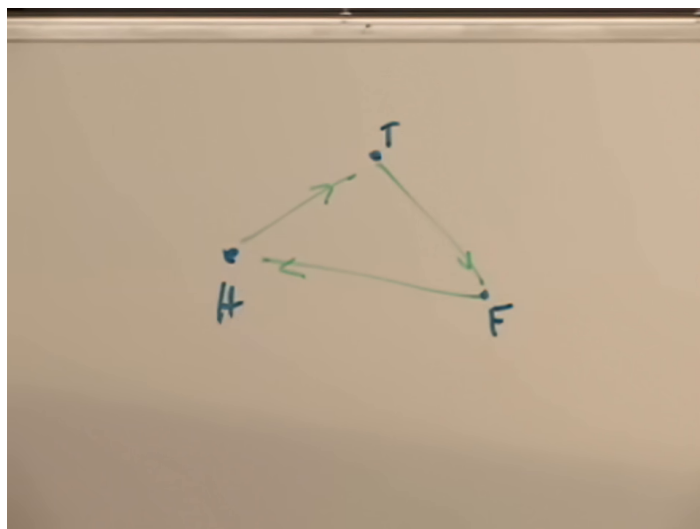


Figure 1.3: The three-state deterministic cycle. Reversing every arrow gives a unique route back from each state.

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Starting from any state, the system cycles forever.

Let the system run for a million time steps, reverse every arrow, and run for another million. It returns to its starting point. This is an operational meaning of reversibility and of information conservation: the present state still contains everything required to reconstruct the past. We need not even know the law in detail; we need only a switch that replaces the evolution by its inverse.

Now add a classical accident. With probability one in a million, suppose the coin stays put when the rule says to move. A short reversal test may succeed, but after ten million steps one or more accidents are likely. The reverse dynamics does not know when the random pauses occurred. In the three-state example the final reverse run is then no more likely to find the original state than either of the other two. Classical randomness has destroyed information.

1.5 An Electron Forward and Backward

Apply the reversal test to a quantum particle. One slit is enough. Send one electron through it, remove the screen, and do not determine where the electron goes. After a time t , reverse the dynamics and let the entire undisturbed system evolve for another time t . In the ideal experiment the electron returns along the reversed evolution and arrives back at the gun. Quantum spreading has not acted like a sequence of independent classical kicks.

Insert an intermediate detector and the result changes. If the detector records where the electron emerged, then reversing the electron's dynamics alone does not erase that record. The exact return test fails. In modern language, the electron has become correlated with degrees of freedom in the detector; the closed combined system may still evolve reversibly, but the electron can no longer be treated as though nothing else carries its history.²

The classical contrast is deliberately mundane. Looking at a coin—the lecture uses a Chilean peso—need not turn heads into tails. If sufficiently energetic light is used, it can of course knock the coin over, but classical physics allows the illumination to be made arbitrarily gentle while still revealing the state. An intermediate classical observation therefore need not spoil the reversal experiment. Quantum measurement admits no analogous limit in which a perfectly sharp observation has an arbitrarily small effect.

The two-slit experiment tells the same story. Interference survives only if nothing records which slit the electron used. The recorder need not be a human observer. A gas molecule, another electron, or any other physical degree of freedom may retain the path information. If the upper path changes one molecule and the lower path changes another, then the alternatives leave distinguishable records. The interference disappears and the observed probabilities reduce to the classical sum in Eq. (1.2).

Question from the audience. If a particle passes through one slit and is deflected, does momentum conservation not give the plate an opposite recoil, thereby recording the path?^a

Response. The recoil is real, but a physical kick is not automatically a readable record. A plate localized well enough to define the slits has an intrinsically broad momentum distribution. The recoil shifts that distribution. If the shift is small compared with its width, the post-collision momentum does not reveal whether the kick occurred.

^aLecture timestamp: 00:44:30–00:44:47.

²Editorial clarification: This closed-system formulation sharpens the lecture's contrast without adding a new formalism.

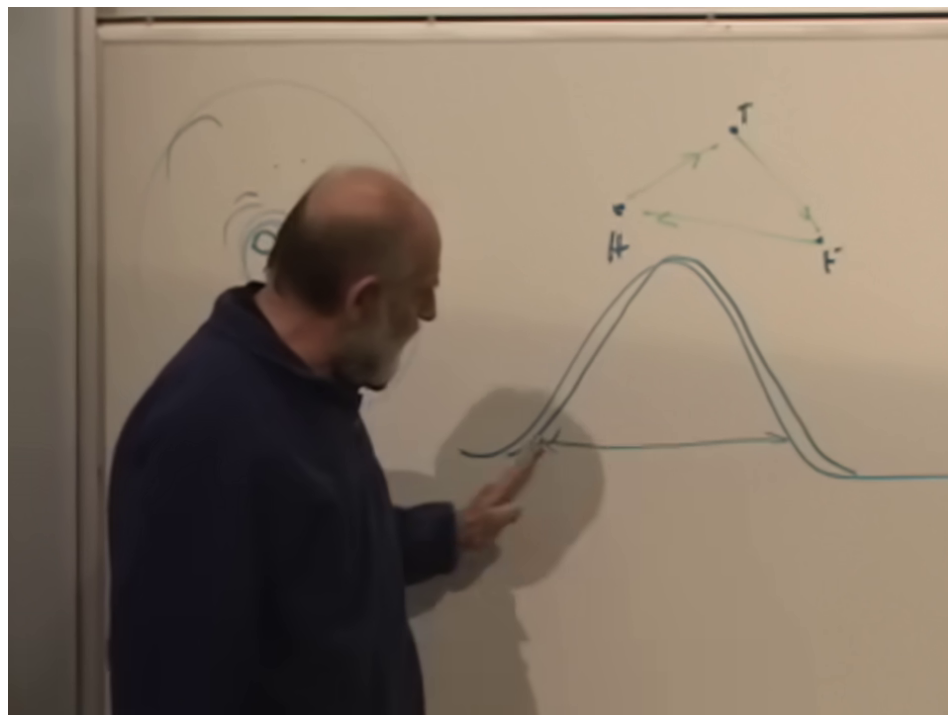


Figure 1.4: The plate's broad momentum profile and its small recoil-induced shift. The two nearly coincident curves are the part of the board relevant to the argument.

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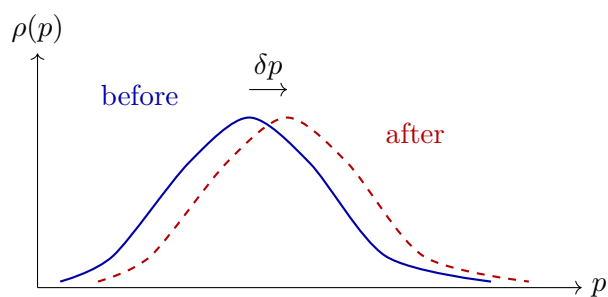


Figure 1.5: The recoil becomes a path record only when the shift δp is distinguishable relative to the initial width.

The uncertainty principle closes the apparent loophole. Making the plate's momentum sharp enough to resolve a tiny kick makes its position correspondingly uncertain, so the slit geometry itself becomes blurred. The apparatus cannot simultaneously define the paths sharply and record an arbitrarily small path-dependent recoil.

1.6 Why Classical Uncertainty Is Not Fundamental

Before deriving the microscope argument, distinguish quantum uncertainty from an ordinary statistical cloud. A Brownian particle is knocked by surrounding molecules, and an ensemble of such particles spreads in position and velocity. In classical physics this uncertainty reflects incomplete observation. With a better microscope, a more accurate detector, and gentler illumination, one imagines determining the position and velocity of each particle as accurately as desired.

Quantum mechanics contains a logical obstruction to that refinement. It is not merely that our apparatus is coarse or expensive. A procedure that makes position sharper necessarily changes what can be said about momentum. Heisenberg first encountered this through abstract mathematics, where the operations corresponding to position and momentum do not commute. The microscope thought experiment supplies the physical illustration.

1.7 Heisenberg's Microscope

To locate a particle optically, scatter light from it and focus the scattered wave into an image. The required chain begins with the energy of one photon,

$$E = hf = \hbar\omega, \quad \omega = 2\pi f, \quad \hbar = \frac{h}{2\pi}. \quad (1.7)$$

Visible light has a frequency of order 10^{15} Hz. Light also carries momentum: an absorbed beam heats a door because it carries energy and can push the door because it carries momentum.

For a nonrelativistic particle,

$$E_{\text{kin}} = \frac{p^2}{2m} = \frac{1}{2}pv, \quad (1.8)$$

whereas for light relativity gives

$$E = cp, \quad p = \frac{E}{c}. \quad (1.9)$$

One cycle has period $T = 1/f$, and in that time a light wave advances by one wavelength. Hence

$$c = \frac{\lambda}{T} = \lambda f, \quad f = \frac{c}{\lambda}. \quad (1.10)$$

Combining this with Eqs. (1.7) and (1.9) gives

$$p_\gamma = \frac{hf}{c} = \frac{h}{\lambda}. \quad (1.11)$$

Short wavelength means large photon momentum.

An image cannot resolve detail much smaller than its wavelength. To locate a particle on a scale Δx , the probe must therefore satisfy, at the order-of-magnitude level,

$$\lambda \lesssim \Delta x. \quad (1.12)$$

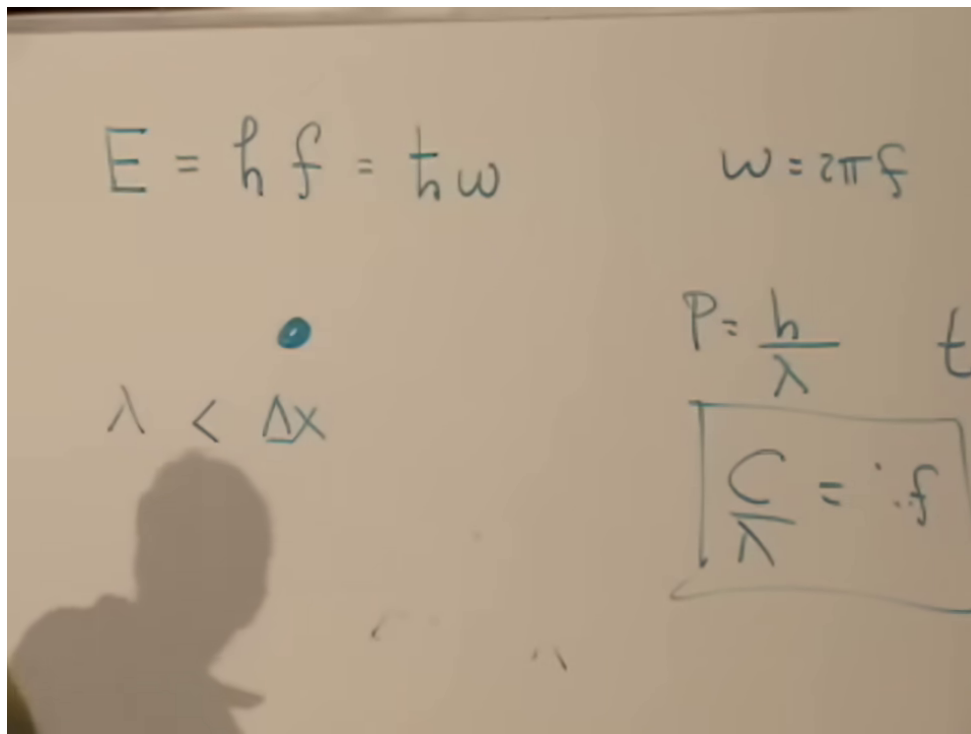


Figure 1.6: The board at the decisive step: $E = hf = \hbar\omega$, $p = h/\lambda$, $c/\lambda = f$, and the imaging condition $\lambda < \Delta x$.

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But the photon used to obtain that resolution carries momentum h/λ . When it scatters from the particle, the unknown scattering direction gives the particle an uncertain kick of the same order:

$$\Delta p_{\text{kick}} \sim \frac{h}{\lambda}. \quad (1.13)$$

Equations (1.12) and (1.13) give the heuristic scale

$$\Delta x \Delta p_{\text{kick}} \gtrsim h \quad (\text{order of magnitude}). \quad (1.14)$$

This is not the later sharp inequality with its convention-dependent numerical factor; it is the microscope logic that motivates it.

Why does the same argument not seem fatal classically? A classical light wave may be divided into arbitrarily small amounts of energy while its wavelength is held fixed. One could imagine forming the same sharp image with an arbitrarily weak wave. Quantum light comes in indivisible packets. At a fixed wavelength the smallest available probe is one photon, and that photon already carries h/λ . A precise position measurement cannot be made arbitrarily gentle. A gentle measurement is possible only by using a long wavelength and accepting a coarse position.

1.8 Measuring Velocity Gently

After the break, a question asks how velocity can be measured at all. The answer is to use two deliberately imprecise position measurements and allow a long time between them. Suppose the particle's mass is known, so that measuring velocity also determines its nonrelativistic momentum.

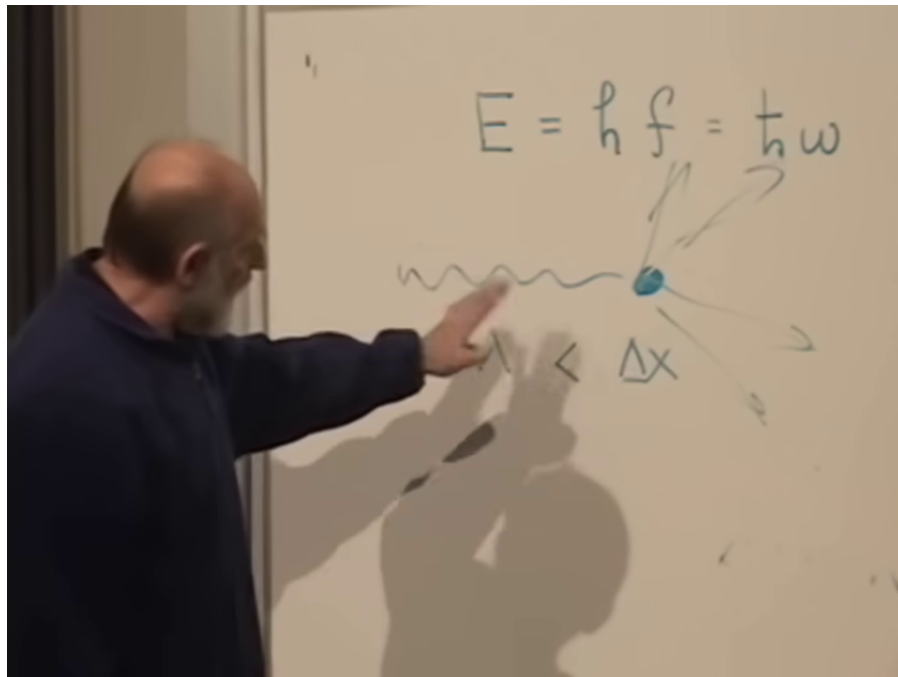


Figure 1.7: A short-wavelength photon localizes the particle sharply, then scatters in an uncontrolled direction and transfers momentum.

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Use a long-wavelength photon for the first observation. It gives only

$$x_1 = x + O(\lambda), \quad (1.15)$$

but its momentum is small enough that the particle receives only a gentle kick. Wait for a time t , then make a second long-wavelength observation:

$$x_2 = x + vt + O(\lambda). \quad (1.16)$$

The inferred displacement and velocity are

$$d = x_2 - x_1 = vt + O(\lambda), \quad (1.17)$$

$$v_{\text{meas}} = \frac{d}{t} = v + O\left(\frac{\lambda}{t}\right). \quad (1.18)$$

No matter how large λ is, the measurement contribution λ/t can be made small by waiting long enough.

The tradeoff has not disappeared. Velocity is determined accurately only while position remains uncertain on the large scale λ . The method does not assign both a sharp position and a sharp momentum at one instant.

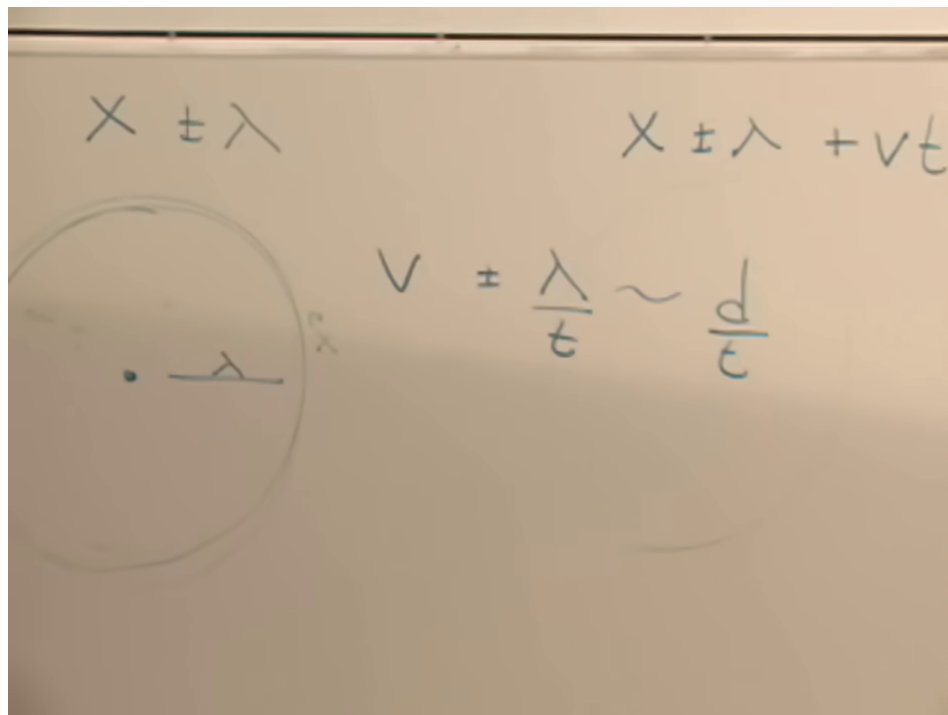


Figure 1.8: The two coarse position readings and the resulting velocity estimate $v = d/t$ with uncertainty of order λ/t .

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Question from the audience. Does the term λ/t include every source of uncertainty in this velocity experiment?^a

Response. No. It describes the uncertainty contributed by the two coarse position readings. The original preparation supplies another uncertainty, and the complete accounting requires keeping both. A fixed reference object such as a wall can have its mean position determined by averaging many gentle photon measurements over a long time. The detailed follow-up is deferred rather than answered by pretending that λ/t is the whole uncertainty principle.

^aLecture timestamp: 01:14:07–01:16:16.

1.9 Classical States Are Points in Phase Space

The experiments point to a failure of classical logic, and the failure appears at the most elementary question: what is a state? For a classical coin the alternatives H and T are points in a set. For a die the set has six points. For one classical particle, a state is a point in phase space,

$$(x, p), \tag{1.19}$$

and for many particles it is the corresponding list of all positions and momenta. Dynamics is a flow carrying one point of this set into another.

If the apparatus is coarse, we may describe our knowledge by a probability density on phase space,

$$\rho(x, p) \geq 0, \quad \int dx dp \rho(x, p) = 1. \tag{1.20}$$

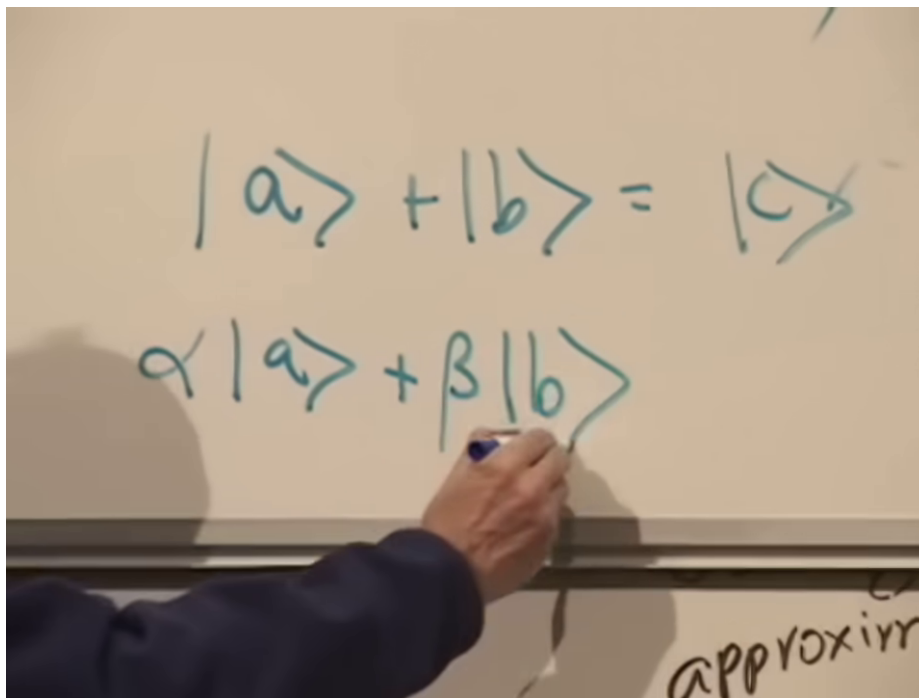


Figure 1.9: Ket-vector addition and a general complex linear combination. The result is another ket vector in the same vector space.

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The distribution may be concentrated in a small region and fall away from its center. Classical logic treats this as incomplete knowledge, not as the maximal possible specification of the state. More delicate, less disturbing measurements can in principle concentrate ρ more closely around one phase-space point.

That conclusion is exactly what quantum mechanics refuses. A quantum state requires more structure than a point-set description supplies. Of course any collection of states can be regarded as a set in the formal mathematical sense; the claim is that set membership and classical probability are not the natural operations that organize quantum alternatives. The state space must support linear combinations.

1.10 Complex Vector Spaces and Ket Vectors

A vector space is a collection of objects called vectors together with two operations. Vectors may be added, and they may be multiplied by scalars. For the spaces used in quantum mechanics, the scalars are complex numbers. Thus if $|a\rangle, |b\rangle \in V$ and $\alpha, \beta \in \mathbb{C}$, then

$$\alpha |a\rangle + \beta |b\rangle \in V. \quad (1.21)$$

This closure under linear combinations is the first mathematical departure from a classical list of alternatives. Multiplying heads by $2 + 4i$ has no meaning in the classical two-point state space; multiplying a ket vector by a complex scalar is a basic operation.

Dirac denoted these vectors by kets such as $|a\rangle$, half of a bracket. The complementary half will be

a bra, and a bra next to a ket makes a bracket. That structure comes later; for the moment the notation only marks an abstract vector.

The word *vector* must not be confused with an arrow in ordinary three-dimensional space. To keep the distinction visible, call the familiar arrow a *pointer*. A pointer has a length and a spatial direction. It can be multiplied by a real number and added to another pointer by the parallelogram rule, so pointers form a vector space over $\mathbb{R}$. Quantum state vectors are more general and form spaces over $\mathbb{C}$.

The lecture occasionally uses *Hilbert space* while introducing this linear structure. Strictly, a complex vector space becomes a Hilbert space only after an inner product and an appropriate completeness condition are supplied. Those additions have not yet been introduced; at this stage we need only the two linear operations.

1.11 Functions as Vectors

Consider complex-valued functions of one real variable,

$$\psi(x) = \psi_R(x) + i\psi_I(x). \quad (1.22)$$

The argument x is real; the value of the function may be complex. Given two such functions and a complex number α , define

$$(\psi + \phi)(x) = \psi(x) + \phi(x), \quad (1.23)$$

$$(\alpha\psi)(x) = \alpha\psi(x). \quad (1.24)$$

Both operations produce new complex-valued functions, so any chosen class of functions closed under these operations forms a complex vector space. Real functions similarly form a vector space over $\mathbb{R}$, but not over $\mathbb{C}$, because multiplying a real function by i takes it out of the real-valued collection.

Function spaces are the examples that motivated Hilbert's work and will become central to wave mechanics. Questions of inner products, normalization, and which functions are admissible are deliberately left for later.

1.12 Columns and Dimension

A finite-dimensional example is a column of n complex numbers. For $n = 4$,

$$A = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}, \quad B = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}. \quad (1.25)$$

Addition and scalar multiplication are componentwise:

$$A + B = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \\ a_4 + b_4 \end{pmatrix}, \quad \alpha A = \begin{pmatrix} \alpha a_1 \\ \alpha a_2 \\ \alpha a_3 \\ \alpha a_4 \end{pmatrix}. \quad (1.26)$$

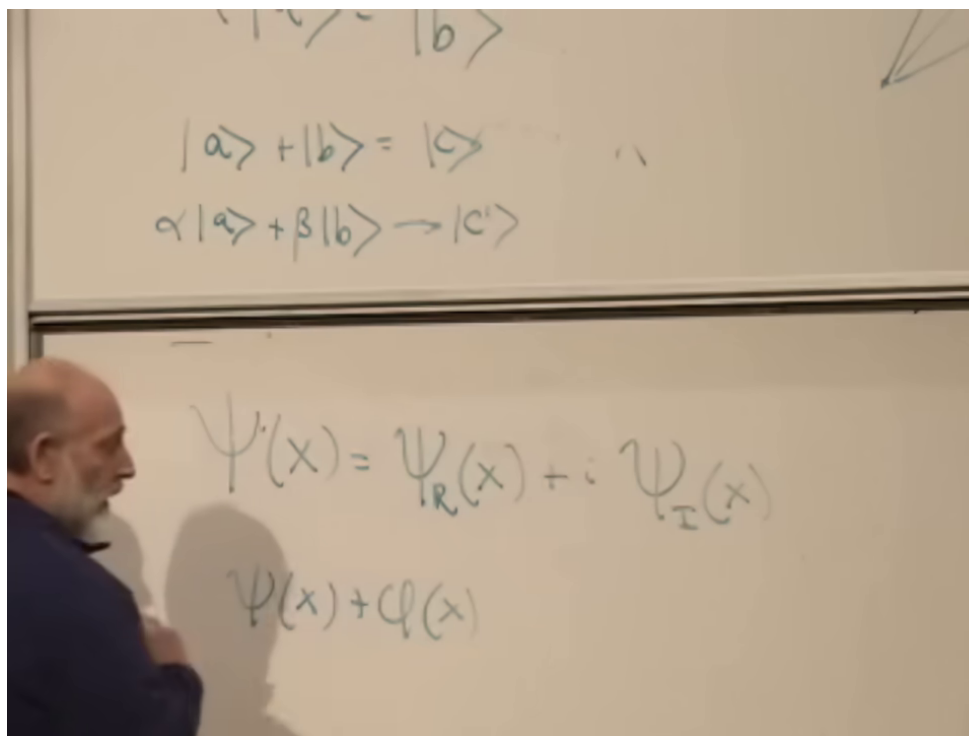


Figure 1.10: A complex function resolved into real and imaginary parts, beneath the ket-vector linear-combination rule.

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No multiplication of one vector by another has been defined. Only addition of vectors and multiplication of a vector by a scalar are needed here.

Columns of a fixed length n form an n -dimensional complex vector space. Pairs, quadruples, and infinite columns each give their own spaces; throwing columns of every possible length into one collection does not give one vector space under these rules, because no addition has been defined between a two-component and a three-component column. The entries may be viewed as components, just as the entries of a spatial pointer are its components along chosen axes. One may also regard a column as a function of its discrete index, but that abstraction is postponed.

We have not yet explained in what physical sense a state is represented by such a vector. The present task is only to learn the operations before applying them to quantum systems.

Question from the audience. At this stage, is the vector-space discussion supposed to sound like mumbo jumbo?^a

Response. Its physical use has not yet been supplied, so the abstraction is expected to feel unmotivated. The definitions themselves are nevertheless precise: vectors can be added and multiplied by complex numbers. The next lectures will connect those operations to physical states.

^aLecture timestamp: 01:44:52–01:45:10.

The analogy with logic is worth retaining. Classical Boolean operations act on sets: *and* corresponds to intersection, while *or* corresponds to union. The lecture momentarily interchanges the two and corrects the order. Quantum logic will be organized by subspaces and linear operations rather

The image shows two equations written on a whiteboard. The top equation illustrates componentwise addition of two four-component column vectors:
$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \\ a_4 + b_4 \end{pmatrix}$$
The bottom equation illustrates complex scalar multiplication of a four-component column vector by a scalar α :
$$\alpha \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} \alpha a_1 \\ \alpha a_2 \\ \alpha a_3 \\ \alpha a_4 \end{pmatrix}$$

Figure 1.11: Componentwise addition and complex scalar multiplication for four-component columns.

Lecture frame: 01:42:00.

than by ordinary unions of alternatives. That is why vector spaces are not optional mathematical decoration.

1.13 Complex Conjugation

The complex numbers themselves form a one-dimensional vector space over $\mathbb{C}$. They also carry an operation that will be indispensable when inner products are introduced. Write

$$z = x + iy. \quad (1.27)$$

Its complex conjugate is

$$z^* = x - iy. \quad (1.28)$$

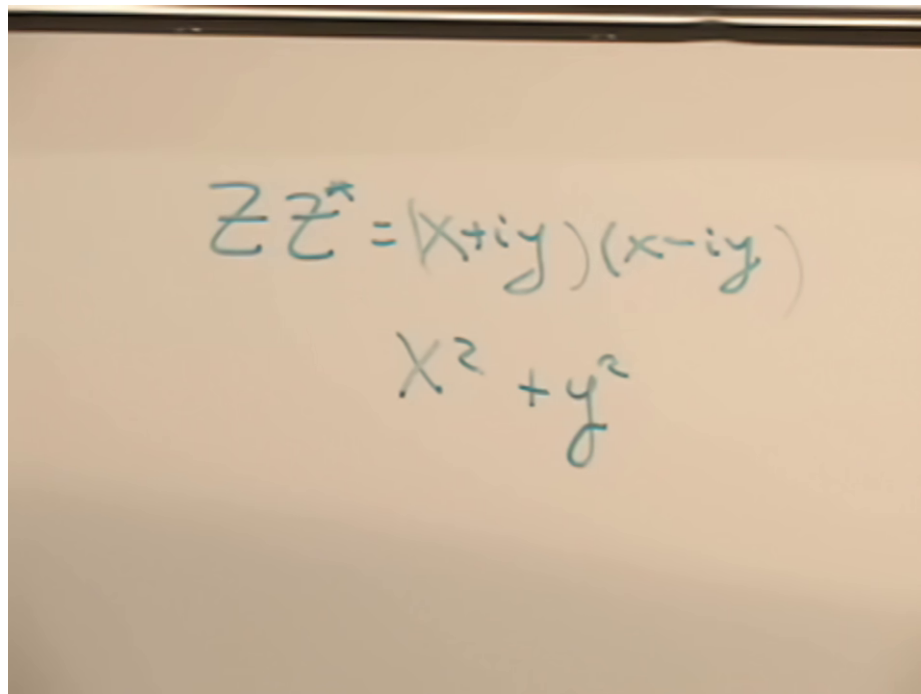
Geometrically, conjugation reflects the point in the complex plane across the real axis. Algebraically,

$$z^* z = (x - iy)(x + iy) \quad (1.29)$$

$$= x^2 + y^2 = |z|^2. \quad (1.30)$$

The cross terms cancel and $i(-i) = 1$, leaving the squared distance from the origin.

Conjugation is best regarded as a map from complex numbers back to complex numbers, not merely as a trick for simplifying products. The next lecture will extend the vector-space language, introduce operators, Hermitian operators, and eigenvalues, and begin applying this machinery to quantum systems.



Handwritten calculation on a board:

$$zz^* = (x+iy)(x-iy)$$
$$x^2 + y^2$$

Figure 1.12: The board calculation $zz^* = (x+iy)(x-iy) = x^2 + y^2$.

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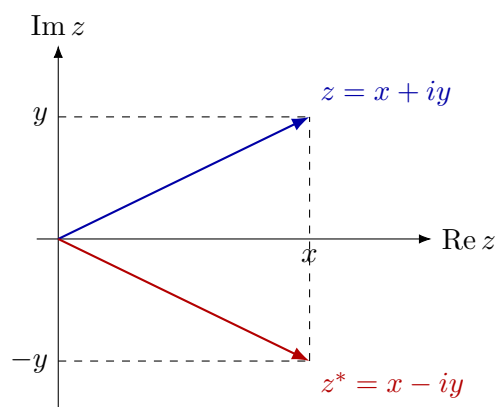


Figure 1.13: Complex conjugation is reflection across the real axis; both vectors have magnitude $\sqrt{x^2 + y^2}$.